Yatin **Dandi**

DUAL DEGREE (BTECH-MTECH) STUDENT · COMPUTER SCIENCE AND ENGINEERING

IIT Kanpur

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Education

Indian Institute of Technology Kanpur — *Kanpur, India*

DUAL DEGREE (BTECH-MTECH), MAJORING IN COMPUTER SCIENCE AND ENGINEERING — *July. 2017 - Present*

- **Cumulative Performance Index : 9.6/10.0**
- **Exchange and visiting student at Ecole polytechnique fédérale de Lausanne (EPFL)**

Interests

Theory of Deep Learning, Optimization, Federated Learning, Decentralized Learning, Causal Inference, Statistical Physics of Computation, Dynamical Systems, Stochastic Differential Equations, Spectral Graph Theory.

Publications and Preprints

Implicit Gradient Alignment in Distributed and Federated Learning — *Virtual Conference*

YATIN DANDI*, LUIS BARBA*, MARTIN JAGGI (* DENOTES EQUAL CONTRIBUTION) — *July, 2021*

FL-ICML 2021 : International Workshop on Federated Learning for User Privacy and Data Confidentiality in Conjunction with ICML 2021. Under review at AAAI, 2022.

Generalized Adversarially Learned Inference — *Virtual Conference*

YATIN DANDI, HOMANGA BHARADHWAJ, ABHISHEK KUMAR, PIYUSH RAI — *February, 2021*

Proceedings of the AAAI Conference on Artificial Intelligence

Jointly Trained Image and Video Generation using Residual Vectors — *Colorado, USA*

YATIN DANDI, ANIKET DAS, SOUMYE SINGHAL, VINAY P. NAMBOODIRI, PIYUSH RAI — *March, 2020*

2020 Winter Conference on Applications of Computer Vision (WACV '20)

Model-Agnostic Learning to Meta-Learn — *Virtual Conference*

ARNOUT DEVOS*, YATIN DANDI* (* DENOTES EQUAL CONTRIBUTION) — *December, 2020*

Pre-registration workshop, NeurIPS (2020). Full paper published in Proceedings of Machine Learning Research (PMLR).

NeurInt-Learning Interpolation by Neural ODEs (Spotlight) — *Virtual Conference*

AVINANDAN BOSE, ANIKET DAS, YATIN DANDI, PIYUSH RAI — *December, 2021*

The Symbiosis of Deep Learning and Differential Equations: DLDE Workshop, NeurIPS 2021. Under review at AAAI, 2022.

Understanding Layer-wise Contributions in Deep Neural Networks through Spectral Analysis

YATIN DANDI, ARTHUR JACOT

Under review at AISTATS 2022

Honors & Awards

2017	**Aditya Birla Scholarship**, Awarded to 15 students from all IITs	*Mumbai, India*
2018,2019	**Academic Excellence Award**, Awarded (twice) for exceptional performace	*IIT Kanpur*
2016	**KVPY Scholarship, 2016**, Indian Institute of Science	*Bangalore*
2015	**NTSE Scholarship**, Government of India	*India*
2017	**All India Rank 135**, JEE Advanced 2017	*India*
2016	**Selected for Indian National Physics and Chemistry Olympiasd**, HBCSE	*Mumbai, India*

Projects and Internships

Data dependent mixing for Faster Heterogeneous Decentralized Learning — *EPFL*

PROJECT UNDER PROF. JAGGI AT THE MLO LAB, EPFL — *August 2021 - Present*

- Extended the analysis for decentralized SGD to characterize the interaction between data heterogeniety and the mixing matrix.
- Formulated a novel criterion for periodically updating the mixing matrix based on an appropriate constrained optimization problem.
- Implemented the approach in toy settings as well as on large scale datasets using the torch.distributed and MPI framework.

Implicit Gradient Alignment in Distributed and Federated Learning
EPFL

Project under Prof. Jaggi at the MLO lab, EPFL — *Apri 2021 - August 2021*

- Studied several theoretical works aiming to explain the generalization benefits of SGD.
- Designed a novel algorithm that replicates the implicit regularization effect of SGD in distributed and federated learning settings.
- Proved theoretical results that show that the proposed algorithm approximates SGD and leads to descent on the regularized objective.
- Demonstrated an analogous implicit regularization effect for SCAFFOLD.

Generalized Adversarially Learned Inference
IIT Kanpur

Undergraduate Project under Prof. Piyush Rai — *October 2019 - May 2020*

- Proposed a novel approach to introduce reconstructive, self-supervised, and knowledge-based feedback in the ALIGAN/BIGAN framework by modifying the discriminator's objective to correctly identify more than just two joint distributions of image-latent vector pairs.
- Performed quantitative and qualitative evaluations over baselines on SVHN and CelebA datasets.
- Proved that the optimum for the proposed objective corresponds to simultaneously matching all the distributions.

Understanding Layer-wise Contributions in Deep Neural Networks through Spectral Analysis
EPFL

Project under Arthur Jacot and Prof. Clement Hongler at EPFL — *March 2020 - May 2021*

- Utilized the properties of Hermite Polynomials and Spherical Harmonics to prove that the eigenvalues corresponding to high frequency functions of the contribution to the NTK from the first layer are larger than those corresponding to the second layers for a two-layer network with arbitrary activation functions.
- Validated the proposed theory using experiments on datasets of Spherical Harmonics as well as MNIST.

Causal Inference, Out of Distribution Generalization and Meta Learning: A Unified Perspective
EPFL

Semester project under Prof. Grossglauser at the INDY lab, EPFL — *August 2020 - December 2020*

- Studied the theory of causal inference, surveyed recent works on causal inference and out of distribution generalization for several classes of machine learning models
- Analyzed the relationship between out of distribution generalization, meta-learning and causal inference
- Formulated and empirically analyzed several algorithms under the novel paradigm of "out of task distribution generalization".

On the Effect of Noise induced by Gradient Stochasticity on Optimizing 2-player Differentiable Games
EPFL

Project under Tatjana Chavdarova at EPFL — *January 2021 - May 2021*

- Analyzed the continuous time limit of stochastic variants of first order algorithms for differentiable games using the theory of Stochastic Differential Equations (SDEs)
- Derived and studied first order approximations for second order algorithms such as SGA (Symplectic Gradient Adjustment) for n-player games.

Towards a Proof of the Fourier Entropy Conjecture
IIT Kanpur

Course Project under Prof. Rajat Mittal — *August 2021 - November 2021*

- Surveyed several papers related to the Fourier Entropy conjecture.
- Prepared a summary report and presentation on the recent work by Kelman et al..

Jointly Trained Image and Video Generation using Residual Vectors
IIT Kanpur

Undergraduate Project under Prof. Piyush Rai and Prof. Vinay Namboodiri — *May 2019 - July 2019*

- Designed a novel modelling technique for joint image and video generation that simultaneously learns to map latent variables with a fixed prior onto real images and interpolate over images to generate videos.
- Implemented Variational Autoencoder and Generative Adversarial Networks based video generation models in Pytorch utilizing the proposed technique.
- Performed quantitative and qualitative evaluations over previous VAE and GAN based video generation approaches.

New York Office, IIT Kanpur
IIT Kanpur

Machine Learning for Large Scale Logistics Platform, under Prof. Manindra Agarwal — *May 2018 - July 2018*

- Implemented a state of the art algorithm for online collaborative filtering based on Fast Matrix Factorization for Online Recommendation with Implicit Feedback (He et al.) using Numpy and improved the model with sentiment and frequency dependent weighting schemes.
- Used Kafka for real-time data processing and simulated interactions using locust.
- Implemented a recommender system based on deep autoencoders and compared the results with other models using metrics such as hit ratio.
- Implemented a Bidirectional LSTM model using Keras for sentiment analysis of user comments.
- Trained the Latent Dirichlet allocation model on Wikipedia articles for automatic extraction of topics.

Social Situation Inference in Simple Animated Shapes
IIT Kanpur

Undergraduate Project under Prof. Nisheeth Srivastava — *May 2018 - July 2018*

- Conducted experiments to characterise the nature and granularity of information about social interactions accessible to human observers from simple visual displays.
- Developed an animation engine capable of producing a vast range of social situations involving autonomous agents having long term and short term goals.

Deep Reinforcement Learning for Atari Games
IIT Kanpur

ASSOCIATION FOR COMPUTING ACTIVITIES, IIT KANPUR *February 2018 - May 2018*

- Used Numpy to implement various reinforcement learning algorithms such as Dynamic Programming (Policy and Value iteration), Monte Carlo (Epsilon-greedy and off-policy), TD Learning (Q-Learning and SARSA) and Q-Learning with Function Approximation.
- Implemented Deep Q-Learning and Policy Gradient methods for Atari Games using PyTorch and OpenAI Gym.

Image Captioning with Visual Attention
IIT Kanpur

PROGRAMMING CLUB, IIT KANPUR *February 2018 - May 2018*

- Compared various CNN based models for image classification and implemented them using eager execution in Tensorflow.
- Implemented the model described in Show, Attend and Tell (Xu et al.2015) using Tensorflow's estimator API and evaluated the model on MS COCO dataset.

Microsoft code.fun.do hackathon
IIT Kanpur

NATIONAL-LEVEL HACKATHON WINNER *February 2018*

- Selected out of 120 students to represent IIT Kanpur at Microsoft Hyderabad center to showcase our project in their academia-industry collaboration event AXLE.
- Built an interactive interface using D3.js to display changing geopolitical relations and popularity of world leaders.
- Used Scrapy, a scraping framework to extract world news.
- Performed sentiment analysis and named entity-recognition on the extracted news to infer the effect of the concerned news on world politics.

Positions of Responsibility

AISTATS, 2022
online

REVIEWER *October 2021 - November 2021*

- Reviewed papers on Matrix Factorization and Adversarial Risk.

Introduction to Machine Learning
IIT Kanpur

TEACHING ASSISTANT *August 2021 - Present*

- Graded assignments and exams for the introductory machine learning course.

SIGML - Special Interest Group on Machine Learning
IIT Kanpur

COORDINATOR *June 2019 - Present*

- Helped conduct and organize talks and lectures on various topics in machine learning.

Programming Club
IIT Kanpur

SECRETARY *April 2018 - April 2019*

- Helped conduct and organize Linux fest, introductory workshops and various hackathons.

Association of Computing Activities
IIT Kanpur

PROJECT MENTOR *January 2019 - April 2019*

- Mentored first-year students on a semester long project on Generative Adversarial Networks.

Programming Club
IIT Kanpur

PROJECT MENTOR *May 2019 - July 2019*

- Mentored first-year students on a semester long project on Probabilistic Machine Learning.

Relevant Courses

Introduction to Programming - A	Probability for Computer Science - A*	Calculus and Real Analysis - A*
Discrete Mathematics - A	Logic for Computer Science - A*	Topics in Probabilistic Modeling and Inference - A
Computational Cognitive Science (audited)	Linear Algebra and ODE - A	Data Structures and Algorithms - A
Bayesian Analysis - A*	Theory of Computation - A	Introduction to Machine Learning - A*
Optimization For Machine Learning - 6.0/6.0	Learning Theory - 6.0/6.0	Statistical Physics of Computation (ongoing)
Mathematical Methods For Physicists	Computational Complexity (ongoing)	Complexity Measures for Boolean Functions (ongoing)
Quantum Computing	Modern Cryptology	Lattice Models (ongoing)

A*: grade for exceptional performance

Programming Skills

Programming Languages	C, C++, Python, Javascript, MATLAB, Haskell
Libraries and frameworks	Tensorflow, Pytorch, Jax, Scikit-Learn, Pillow, Keras, Numpy
Web	Flask, HTML, CSS, jQuery, MySQL
Utilities	Linux shell utilities, LaTeX, Git, Docker

Miscellaneous

- Delivered a talk on Zero Knowledge Proofs for the course Complexity Theory, IIT Kanpur and on its applications to machine learning at the MLO lab, EPFL.
- Delivered a talk on Brain-Computer Interface in an event organized by Science Coffee House, IIT Kanpur.
- Winner of Blackbox - a three hour high speed hackathon based on an Esoteric language organized by Programming Club IIT Kanpur.